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Paleomagnetism of the Middle–Late Jurassic to Cretaceous red beds from the Peninsular Thailand: Implications for collision tectonics

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ABSTRACT

Jurassic to Cretaceous red sandstones were sampled at 33 sites from the Khlong Min and Lam Thap formations of the Trang Syncline (7.6°N, 99.6°E), the Peninsular Thailand. Rock magnetic experiments generally revealed hematite as a carrier of natural remanent magnetization. Stepwise thermal demagnetization isolates remanent components with unblocking temperatures of 620–690 °C. An easterly deflected declination ($D = 31.1°$, $I = 12.2°$, $\alpha_{95} = 13.9°$, $N = 9$, in stratigraphic coordinates) is observed as pre-folding remanent magnetization from North Trang Syncline, whereas westerly deflected declination ($D = 342.8°$, $I = 22.3°$, $\alpha_{95} = 12.7°$, $N = 13$ in geographic coordinates) appears in the post-folding remanent magnetization from West Trang Syncline. These observations suggest an occurrence of two opposite tectonic rotations in the Trang area, which as a part of Thai–Malay Peninsula received clockwise rotation after Jurassic together with Shan-Thai and Indochina blocks. Between the Late Cretaceous and Middle Miocene, this area as a part of southern Sundaland Block experienced up to 24.5° ± 11.5° counter-clockwise rotation with respect to South China Block. This post-Cretaceous tectonic rotation in Trang area is considered as a part of large scale counter-clockwise rotation experienced by the southern Sundaland Block (including the Peninsular Malaysia, Borneo and south Sulawesi areas) as a result of Australian Plate collision with southeast Asia. Within the framework of Sundaland Block, the northern boundary of counter-clockwise rotated zone lies between the Trang area and the Khorat Basin.

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1. Introduction

The Sundaland Block of SE Asia comprises the Indochina Peninsula, Thai–Malay Peninsula, Sumatra, Java, Borneo and a shallow marine shelf "*the Sunda Shelf*" (Hall and Morley, 2004; Hall et al., 2008). This block, which is bounded to southwest by a consuming boundary of the Indo-Australian Plate, is located between two well known convergent zones, i.e., the India–Asia collision zone in the northwest and the Australia–Asia collision zone in the southeast (Fig. 1).

Over the past three decades, deformational features of the Asian Continent have been studied in detail around the eastern Himalayan Syntaxis, where the Indian Continent collided with Asia at about 50 Ma. As shown in Fig. 1, clockwise (CW) rotational pattern has been observed around eastern Himalayan Syntaxis by GPS measurements (Wang et al., 2001; Zhang et al., 2004; Gan et al., 2007; Allmendinger et al., 2007; Simons et al., 2007; Copley, 2008) as well as by paleomagnetic investigations (e.g. Dupont-Nivet et al., 2004; Tanaka et al., 2008). Features of clockwise rotational deformation have also been observed in the Shan-Thai Block, the northern Sundaland Block (Simons et al., 2007; Tanaka et al., 2008) and the Indochina Block (Yang and Besse, 1993; Takemoto et al., 2009).

However, paleomagnetic data reported from southern Sundaland (Fig. 1) reveal counter-clockwise (CCW) deflection in declination (Richter et al., 1999; Fuller et al., 1999). Similar trend of declination has been observed in the Mesozoic and Cenozoic rocks of Borneo, South Sulawesi, Celebes Sea, and Peninsular Malaysia (Haile, 1978; Sasajima et al., 1980; Schmidtke et al., 1990; Shibuya et al., 1991; Richter et al., 1999; Fuller et al., 1999). This divergence in paleomagnetic declinations between northern and southern parts of Sundaland indicate an occurrence of internal tectonic deformation in this block, which can be taken as an important clue to understand the effects of deformation from two different collisional regimes (Australia vs. SE Asia and India vs. Asia).

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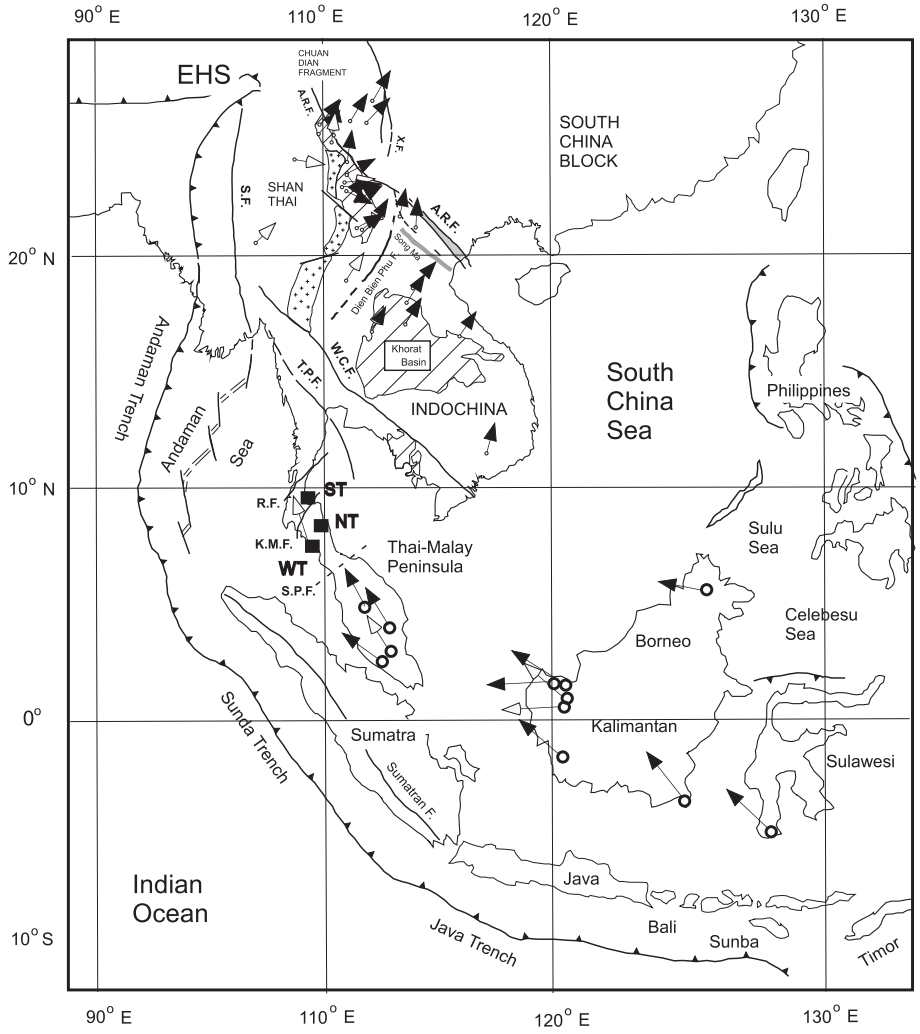


Fig. 1. A simplified tectono-geographical map of southeast-Asia showing the positions of main sutures and faults (modified from Hall, 2002; Hall et al., 2008). In Indochina peninsula, the shaded areas indicate distribution of the Jurassic to Cretaceous red beds and the dotted zone represents the Chongshan–Lancang–Chiang Mai Belt. Sampling sites are indicated by closed squares (ST; Surat Thani; NT; the North Trang Syncline, WT; the West Trang Syncline). Arrows represent the observed declinations at studied localities from Indochina Peninsula, including the Chuan Dian Fragment, the Shan-Thai and Indochina blocks (Takemoto et al., 2009) and from southern Sundaland Block (as listed in Table 2), where Jurassic is marked by open arrow and Cretaceous by solid arrow. A.R.F., Ailao Shan-Red River Fault; E.S.H., Eastern Himalayan Syntaxis; K.M.F., Klong Marui fault; R.F., Panong Fault; S.F., Sagaing Fault, S.C.F., Song Ca Fault; S.P.F, Inferred Songkhla-Penang Fault; T.P.F., Three Pagodas Fault; W.C.F., Wang Chao Fault; X.F., Xianshuihe Fault.

In order to document a tectonic boundary between the CCW and CW rotated areas of the Sundaland Block, we present new paleomagnetic results from Peninsular Thailand. For this purpose, oriented samples were collected from Surat Thani (9.1°N) and Trang (7.6°N) areas, located near the morphologic/tectonic bend of the Sundaland Block (Fig. 1).

2. Geological settings and sampling

The Peninsular Thailand is considered as a part of the Sibumasu Block, which rifted from northern Gondwanaland and eventually collided with Indochina and East Malaysian blocks during the Middle–Late Triassic (Metcalfe, 2002; Ferrari et al., 2008; Barber and Crow, 2009). A belt of Mesozoic continental red beds extends from Myanmar through the Peninsular Thailand into Peninsular Malaysia. A non-marine red beds unit, named as the Trang Group, forms part of this belt in southern Thailand (Meesook et al., 2002).

The Trang Group, which was deposited in brackish and freshwater environment during the Middle–Late Jurassic to Late Cretaceous (Meesook et al., 2002), is divided into four formations in ascending order: the Khlong Min Formation; the Lam Thap Formation; the Sam Chom Formation and the Phun Phin Formation. The Khlong Min Formation contains interbeds of fossiliferous limestone, mudstones and siltstones with some intercalation of biomicrite. The Lam Thap Formation includes thickly-bedded sandstones and siltstones. Based on some fossil findings, i.e., *palynomorphs*, *charophytes* and *turtles* (Lei, 1993; Buffetaut et al., 1994; Tong et al., 2002), Middle to Late Jurassic age has been assigned to the Khlong Min Formation (Buffetaut et al., 2005). Following the formation of a gentle syncline with NE–SW to N–S trending fold axes in the Trang Group, deposition of the Tertiary sedimentary units took place (Chaodumrong and Chaimanee, 2002).

As shown in Fig. 2 (from DMR, 1999), the Middle–Late Jurassic to Late Cretaceous Khlong Min and Lam Thap formations are exposed in the study area around Surat Thani (9.1°N, 99.4°E) and Trang (7.6°N, 99.6°E) localities. By their pioneering work, Bhongsuwan and Ponathong (2002) reported paleomagnetic results from only one site of the Lam Thap Formation; we thus focused our attention on this formation and collected oriented

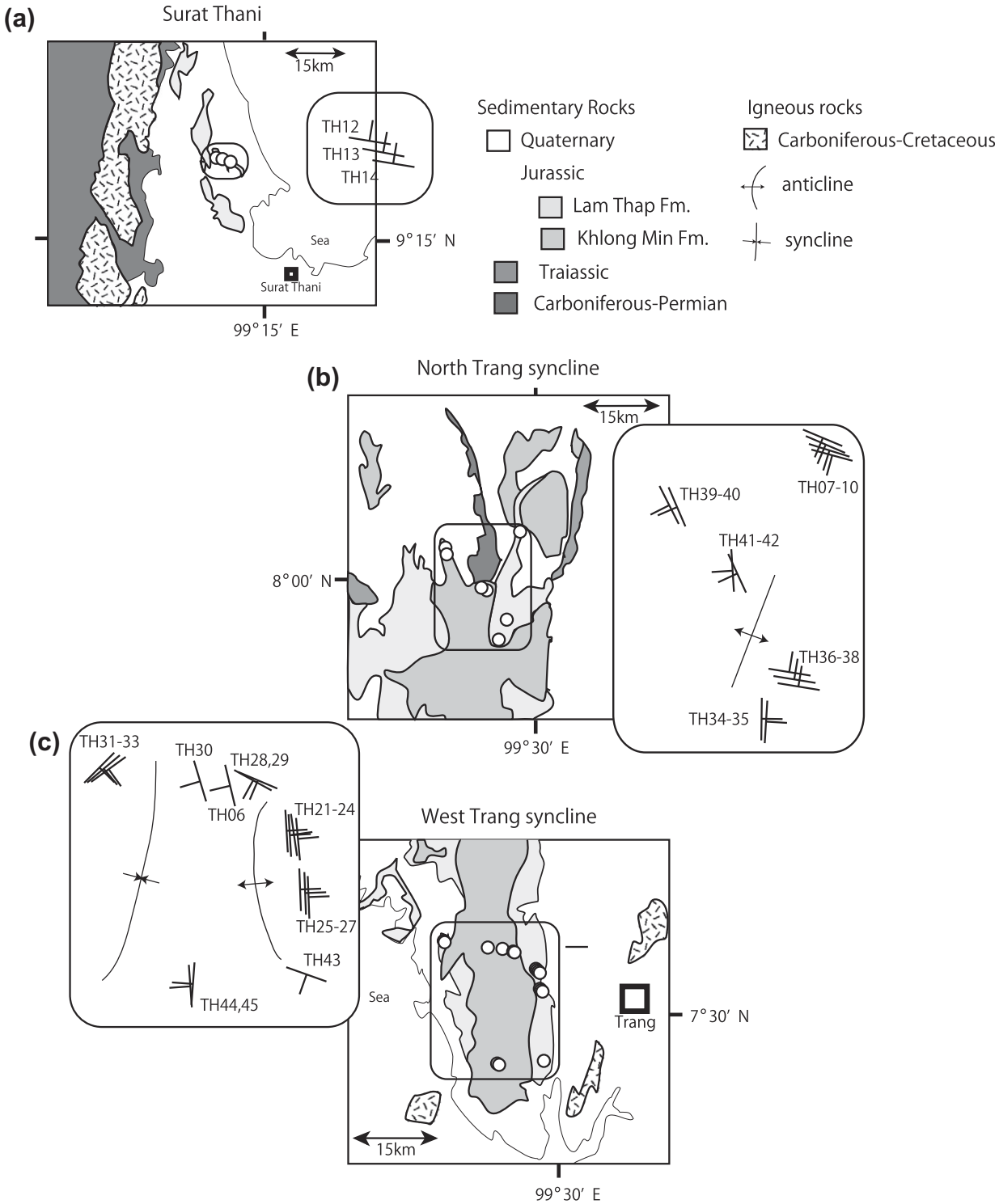


Fig. 2. Geological map of the study area (after DMR, 1999). (a) The Surat Thani locality, (b) the North Trang Syncline and (c) the West Trang Syncline. Sampling sites are shown by closed circles. Strike and dip of strata at each sampling site are shown in the insets.

samples from red sandstone and siltstone at 12 sites. Further 21 sites were sampled from Khlong Min Formation. Sampling localities are distributed in the North Trang Syncline, the West Trang Syncline and the Surat Thani area.

Generally, about 10 oriented hand samples were collected at each site using magnetic compass. Locations of the sampling sites were determined with portable GPS equipment (EMPEX GPS PLUS). The present declination value at each sampling site is determined from the International Geomagnetic Reference Field (Macmillan and Maus, 2005).

3. Laboratory procedures

In the laboratory, one to six cylindrical specimens (25 mm in diameter and 23 mm in height) were prepared from each sample.

Natural remnant magnetizations (NRMs) of all specimens were measured using 2G Enterprises cryogenic magnetometer. Stepwise thermal demagnetization was performed up to 690 °C using Natsuhara TDS-1 thermal demagnetizer, with residual field of less than 5 nT. Temperature increment of 50 °C, 30 °C and 10 °C were used from 100 to 500 °C, 530 to 650 °C and 650 to 690 °C, respectively. NRMs were plotted on orthogonal vector diagrams (Zijderveld, 1967) to assess component structure as well as on equal-area projections to evaluate directional stability. Principal component analysis (Kirschvink, 1980) was used to determine directions of the remanent magnetizations. Mean directions were calculated using Fisherian statistics (Fisher, 1953). The fold test of McFadden's (1990) was calculated using a program package of Cogné (2003).

In order to determine coercivity spectra of the ferromagnetic minerals, progressive acquisition of isothermal remanent magnetization (IRM) up to 2.7 T were performed using 2G pulse magnetizer. In case of composite IRMs, hard, medium and soft components were induced by the DC fields of 2.7 T, 0.4 T, and 0.12 T, respectively. Thermal demagnetization of the composite IRMs (Lowrie, 1990) was carried out to determine the unblocking temperature spectra. Thermomagnetic analyses were conducted on several specimens in air using an automatic recording thermomagnetic balance of the Doshisha University, with magnetic field of 0.3 T.

4. Rock magnetism

Rock magnetic analyses of the red bed samples from Khlong Min and Lam Thap formations revealed hematite as a dominant magnetic carrier (Fig. 3). The IRM acquisition curves indicate that no saturation is achieved up to a maximum field of 2.7 T, revealing

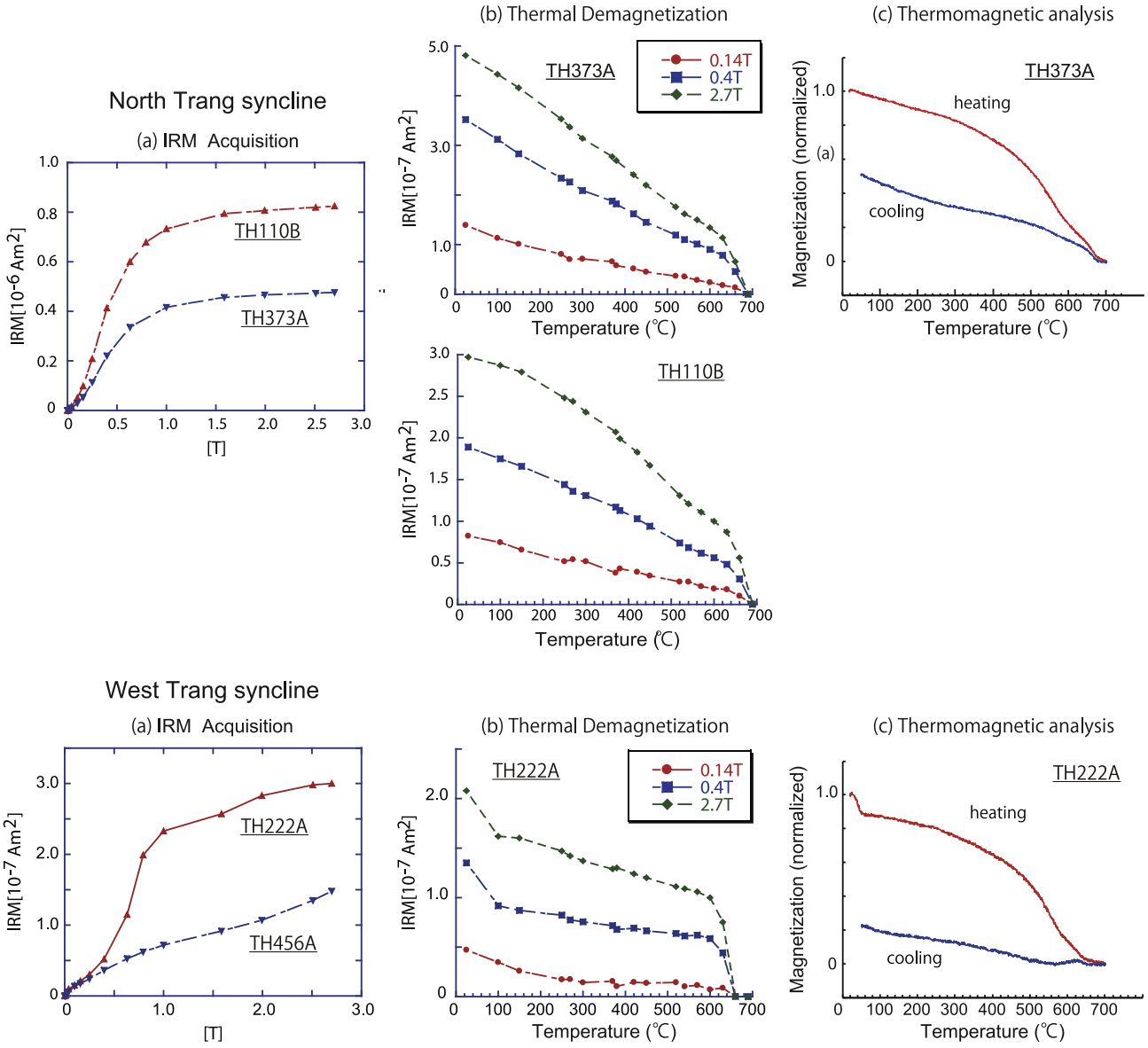


Fig. 3. IRM acquisition and thermal demagnetization of the composite IRMs (imparted with DC fields of 2.7 T, 0.4 T and 0.12 T along three perpendicular axes) are shown in the right box. Thermo-magnetic analysis of the representative specimens from North Trang Syncline (site TH10-sample TH110B and site TH37-sample TH373A) and West Trang Synclines (site TH22-sample TH222A and site TH45-sample TH456A) are also shown in Figure. As indicated, even up to a maximum field of 2.7 T, the IRM saturation point is not achieved. Thermal demagnetization of the hard IRM component places the unblocking temperatures of specimens from North and West Trang Synclines at 690 °C and 660 °C, respectively. Thermo-magnetic analysis of the representative specimens also shows Curie point of about 660 °C for West Trang Syncline, which is lower than that obtained from North Trang Syncline (680 °C).

Table 1
Paleomagnetic results of the red bed samples from middle-upper Jurassic to upper Cretaceous formations: (a) the North Trang syncline, (b) the West Trang syncline and (c) the Surat Thani locality. Magnetization directions for high and intermediate temperature components are listed in this Table. Lat. and Long. are for latitude and longitude, respectively. Bedding attitude is marked by strike and dip angle. *N* and *n* are number of samples measured and used for calculation, respectively. Dec. and Inc. are declination and inclination, respectively; *k* is the Fisherian precision parameter (Fisher, 1953); α_{95} is the radius of cone at 95% confidence level about the mean direction. Sites with asterisk (*) are excluded from mean direction calculation. 0% and 100% are parameters in geographic and stratigraphic coordinates, respectively.

Site	Locality				Formation	NRM intensity (mA/m)	Component	UBT (°C)	*n*/*N*	In-situ		Tilt-corrected		*k*	α_{95}
	Lat. (°N)	Long. (°E)	Strike (°)	Dip (°)						Dec.(°)	Inc.(°)	Dec.(°)	Inc.(°)		(°)
(a) Site and formation mean directions from the North Trang Syncline															
[High-temperature component]															
TH07	8.09	99.45	140	18	Khlong Min	3.9 ± 2.2	Three	690	9/10	17.7	−5.5	17.3	9.4	32.9	9.1
TH08	8.09	99.45	144	27	Khlong Min	3.4 ± 1.7	Three	690	8/9	46.7	−2.2	46.1	24.5	13.6	15.5
TH09	8.09	99.45	130	15	Khlong Min	2.1 ± 0.4	Three	690	10/10	34.1	−8.5	34.2	6.3	8.4	17.7
TH10	8.09	99.45	140	18	Khlong Min	2.4 ± 1.1	Three	690	8/10	46.7	2.9	47.3	17.6	136.2	4.2
TH39*	8.06	99.37	162	49	Lam Thap	1.0 ± 0.3	Single	500	5/10	25.4	11.7	2.5	39.8	15.7	19.9
TH40	8.06	99.37	168	54	Lam Thap	1.2 ± 0.6	Two	6–10	9/9	11.6	−10.7	70.0	42.6	20.6	11.6
TH41*	7.97	99.42	171	37	Lam Thap	5.8 ± 1.3	Single	670	8/8	9.6	30.2	345.5	35.0	560.2	2.3
TH42*	7.97	99.42	149	34.5	Lam Thap	3.2 ± 0.8	Single	680	9/10	7.2	14.1	354.4	33.2	132.5	4.8
TH36	7.92	99.43	289	21	Khlong Min	5.1 ± 0.7	Two	680	9/9	29.4	37.3	27.4	16.7	41.8	8.1
TH37	7.92	99.43	284	17.5	Khlong Min	5.7 ± 2.0	Two	680	8/10	38.9	37.7	34.8	21.7	94.5	5.7
TH38	7.92	99.43	280	18.5	Khlong Min	3.8 ± 0.7	Two	680	10/10	67.3	37.8	57.7	26.6	36.4	8.1
TH34	7.90	99.41	359	84	Khlong Min	10.2 ± 2.0	Two	690	10/10	10.3	5.7	5.6	−10.9	180.1	3.6
TH35	7.90	99.41	2.5	86	Khlong Min	32 ± 56	Two	690	9/10	9.0	12.0	13.7	−6.9	30.6	9.5
Formation mean direction															
(1) Khlong Min Formation (high-temperature component for sites with multi-components magnetization)															
0%	8.0	99.4							9/9	32.5	13.5			9.5	17.6
100%									9/9			31.1	12.2	14.6	13.9
(2) Khlong MingFormation + TH40															
0%	8.0	99.4							10/10	36.5	11.2			8.1	18.1
100%									10/10			34.1	15.6	11.0	15.2
[Intermediate temperature component: 200 °C–570 °C]															
TH07	8.09	99.45	140	18	Khlong Min		Three		10/10	301.8	−20.5	309.2	−25.3	7.8	18.5
TH08	8.09	99.45	144	27	Khlong Min		Three		7/9	288.3	−6.8	294.1	−21.6	4.1	33.9
TH09	8.09	99.45	130	15	Khlong Min		Three		10/10	259.1	−21.3	264.3	−32.5	27.8	9.3
TH10	8.09	99.45	140	18	Khlong Min		Three		10/10	256.9	−19.8	261.5	−31.4	29.1	9.1
Formation mean direction															
0%	8.1	99.4							4/4	276.6	−18.0			13.5	25.8
100%									4/4			282.9	−29.2	15.0	24.6
(b) Site and formation mean directions from the West Trang Syncline															
[High-temperature component: 200 °C–690 °C]															
TH31	7.62	99.32	46	89	Khlong Min	1.7 ± 0.3	Two	680	10/10	318.8	50.2	133.0	39.9	42.4	7.5
TH32	7.62	99.32	41	85	Khlong Min	0.80 ± 0.22	Two	680	5/10	352.9	30.1	78.2	42.9	165.0	6.0
TH33*	7.62	99.32	33	83	Khlong Min	0.96	–	500							
TH06	7.55	99.46	154	60	Khlong Min	3.4 ± 1.0	Single	670	10/10	330.9	−1.8	334.3	−3.2	20.0	11.1
TH21	7.55	99.48	357	53	Khlong Min	12.1 ± 9.9	Single	670	9/9	1.8	7.4	6.3	2.0	62.0	6.6
TH22	7.55	99.48	351	49	Khlong Min	4.6 ± 2.4	Single	670	9/9	1.8	9.2	5.1	−0.2	326.3	2.9
TH23*	7.55	99.48	347	44	Khlong Min	1.4 ± 1.0			590						
TH28	7.55	99.46	127	53	Lam Thap	1.4 ± 0.7	Single	670	10/10	339.2	1.0	326.9	25.7	20.3	11.0
TH29	7.55	99.46	123	43.5	Lam Thap	3.4 ± 1.6	Single	670	9/10	324.1	7.6	315.0	20.5	77.2	11.0
TH30*	7.55	99.41	161	43	Lam Thap	0.81 ± 0.09	–	500							

TH24	7.54	99.48	347	44	Khlong Min	5.9 ± 3.4	Single	670	8/8	358.3	9.8	1.6	−0.3	178.5	4.2
TH25	7.54	99.48	356	52	Khlong Min	1.5 ± 0.3	Single	670	10/10	322.2	37.9	13.9	46.0	18.8	11.4
TH26	7.54	99.48	353	57	Khlong Min	1.3 ± 0.1	Single	670	9/9	328.8	23.5	2.0	31.9	24.8	10.5
TH27	7.53	99.48	353	64	Khlong Min	2.1 ± 0.7	Single	670	10/10	316.6	39.9	24.0	43.1	36.1	8.2
TH43*	7.45	99.48	139	47	Khlong Min	0.93 ± 0.35	–	500							
TH44	7.44	99.35	179	26.5	Khlong Min	2.1 ± 1.3	Single	670	10/10	13.6	31.5	356.5	34.4	224.2	3.2
TH45	7.44	99.35	185	28	Khlong Min	2.4 ± 0.8	Two	690	9/10	351.5	29.1	339.2	31.2	31.2	9.4
Formation mean direction (High-temperature component)															
0%									13/13	342.8	22.3			11.6	12.7
100%	7.5	99.4										0.2	29.1	4.2	23.1
(C) Site and formation mean directions from the Surat Thani locality															
[High-temperature component]															
TH12	9.45	99.16	266	20	Lam Thap	6.2 ± 5.1	Single	690	8/10	9.8	22.7	8.8	3.4	40.1	8.9
TH13	9.45	99.16	268	21	Lam Thap	1.3 ± 1.3	Two	620	10/10	20.6	19	19.3	−0.3	12.3	14.4
TH14	9.45	99.16	274	34	Lam Thap	1.5 ± 1.4	Two	620	10/10	3.1	15.3	3.3	−17.9	8.1	18.0
Formation mean direction															
0%								3/3	11.1	19.0			78.4	14.0	
100%	9.5	99.1						3/3			10.6	−4.9	33.9	21.5	

the presence of high-coercivity minerals, such as hematite or goethite. The presence of hematite is also indicated by unblocking temperature level (between 660 °C and 690 °C) of the hard component during thermal demagnetization of the composite IRMs. Samples from North Trang Synclines give an unblocking temperature level of 690 °C, whereas samples from West Trang Syncline show relatively lower unblocking temperature (about 660 °C). Almost similar tendency is observed in thermo-magnetic analysis, where Curie temperatures of 680 °C and 660 °C have been observed in samples from North Trang and West Trang Synclines, respectively.

Trace amount of goethite is found in the red beds from West Trang Syncline. The unblocking and Curie temperatures of about 100 °C are indicated in the composite IRMs (hard and intermediate components) and thermomagnetic curves, indicating the presence of goethite in the studied rocks.

The presence of magnetite is suspected in samples from four sites (TH07, TH08, TH09 and TH10) of the North Trang Synclines (e.g. TH110B in Fig. 3), where inflection appears at 580 °C in hard coercivity fraction (2.7 T) of the composite IRMs during thermal demagnetization.

5. Paleomagnetic results

5.1. Demagnetization behavior

As listed in the Table 1, initial NRM intensities in the studied specimens vary between 0.6 and 24 mA/m for the Lam Thap Formation and between 0.3 and 199 mA/m for the Khlong Min Formation. Due to weak intensity (less than 1.4 mA/m) and erratic demagnetization behavior after 500 °C, no reliable magnetization direction could be separated from four sites (TH23, TH30, TH33 and TH43) of the West Trang Syncline by principal component analysis.

Almost uni-vectorial magnetization appeared in five sites of the Lam Thap Formation (Fig. 4a, e, h) and seven sites of the Khlong Min Formation from West Trang Syncline (Fig. 4f). Two-component behavior is observed in three sites (TH13, TH14 and TH40) of the Lam Thap Formation and eight sites of the Khlong Min Formation (Fig. 4b, c, g). In this category, low-temperature component is generally erased by 350 °C, whereas high-temperature component appears between 400 °C and 690 °C. In both categories of sites, characteristic component is unblocked between 620 °C and 690 °C. The unblocking temperatures in the samples of North Trang Syncline, West Trang Syncline and the Surat Thani area are characterized by 690 °C, 670 °C, and 620 °C, respectively.

The remaining four sites (TH07, TH08, TH09 and TH10) from North Trang Syncline show three components behavior (Fig. 4d). After the removal of low-temperature component by 350 °C, an intermediate-temperature component appeared between 350 °C and 590 °C, while high-temperature component is observed between 590 °C and 690 °C. The presence of magnetite together with hematite is expected from paleomagnetic and rock magnetic analyses.

5.2. Paleomagnetic directions

Mean direction calculated for low-temperature component is $D = 354.5°$, $I = 12.2°$, $\alpha_{95} = 6.5°$ for the North Trang Syncline, $D = 354.7°$, $I = 18.7°$, $\alpha_{95} = 9.5°$ for the West Trang Syncline and $D = 359.3°$, $I = 7.5°$, $\alpha_{95} = 7.7°$ for the Surat Thani area (Fig. 5). In geographic coordinates, these directions appears almost identical either to the axial geomagnetic dipole field ($D = 0°$, $I = 18.5°$) or the present geomagnetic field direction in the study area ($D = 359.5°$, $I = 3.4°$; Macmillan and Maus, 2005). An unblocking temperature of about 350 °C for the low-temperature component

suggests a VRM carried by hematite as explained by blocking diagrams (Pullaiah et al., 1975).

5.2.1. The North Trang Syncline

High-temperature component is isolated from nine sites in the Khlong Min Formation and four sites in the Lam Thap Formation. Three sites from Lam Thap Formation, which carry single-component magnetization, reveal anomalous behavior. Tilt-corrected mean direction from these three sites ($D = 356.0°$, $I = 36.2°$, $\alpha_{95} = 11.6°$, $N = 3$) show significant deviation with respect to that from 10 sites of the multi-components magnetization. As suggested by different authors (Schmidtke et al., 1990; Deng et al., 2007), a tropical weathering environment probably brought remagnetization to these sedimentary rocks. In order to remove the effects of remagnetization from our data, only high-temperature component is used for mean direction calculation.

Mean direction calculated for nine sites of the Khlong Min Formation (Fig. 6a) gave an easterly deflected declination and shallow positive inclination ($D = 32.5°$, $I = 13.5°$, $\alpha_{95} = 17.6°$ in geographic coordinates, $D = 31.1°$, $I = 12.2°$, $\alpha_{95} = 13.9°$ in stratigraphic coordinates). The DC-tilt test appears positive at 95% confidence limit (Enkin, 2003), where an optimal concentration reaches upon 65.8 ± 40.2% un-tilting. Following the fold test of McFadden's (1990), a value (ξ_1) of 5.064 is obtained in the geographic coordinates and 2.864 in the stratigraphic coordinates, while critical value (ξc) of 3.497 is obtained at 95% confidence level. However, with the addition of one site from Lam Thap Formation (TH40), tilt-corrected mean direction of $D = 34.1°$, $I = 15.6°$, $\alpha_{95} = 15.2°$ is calculated for 10 sites, indicating no significant change in directional and clustering parameters. Although, the DC-tilt test from Enkin (2003) appears undetermined, the McFadden's Fold test proves positive at 95% confidence level. Based on these evidences, a pre-folding origin (Middle–Late Jurassic to Cretaceous) is assigned to this characteristic component from North Trang Syncline.

Although, numerical analysis suggests pre-folding origin for this high-temperature component, directional behavior shows planar distribution rather than Fisherian. This may be due to unsuccessful isolation of low-temperature component in the Lam Thap Formation or un-complete structural correction of the synclinal attitude. Additional paleomagnetic data are required from this syncline to properly address these issues. As a first order approximation, mean direction of nine sites at 100% un-tilting from Khlong Min Formation ($D = 31.1°$, $I = 12.2°$, $\alpha_{95} = 3.9°$) is recognized as the Middle to Late Jurassic characteristic direction for North Trang Syncline, because both the fold tests (Enkin, 2003; McFadden, 1990) are passed. A corresponding paleomagnetic pole for this characteristic direction is 59.1°N, 190.9°E with $A_{95} = 10.1°$. Almost identical easterly deflected declination has been reported as a preliminary paleomagnetic study ($D = 32.6°$) from Lam Thap Formation, although, only one site has been used for this estimation (Bhongsuwan and Ponathong, 2002).

An intermediate-temperature component is derived from only four sites (TH07, TH08, TH09, and TH10) of the Khlong Min Formation, which gave a mean direction of westerly deflected declination and upward inclination ($D = 276.6°$, $I = -18.0°$, $\alpha_{95} = 25.8°$

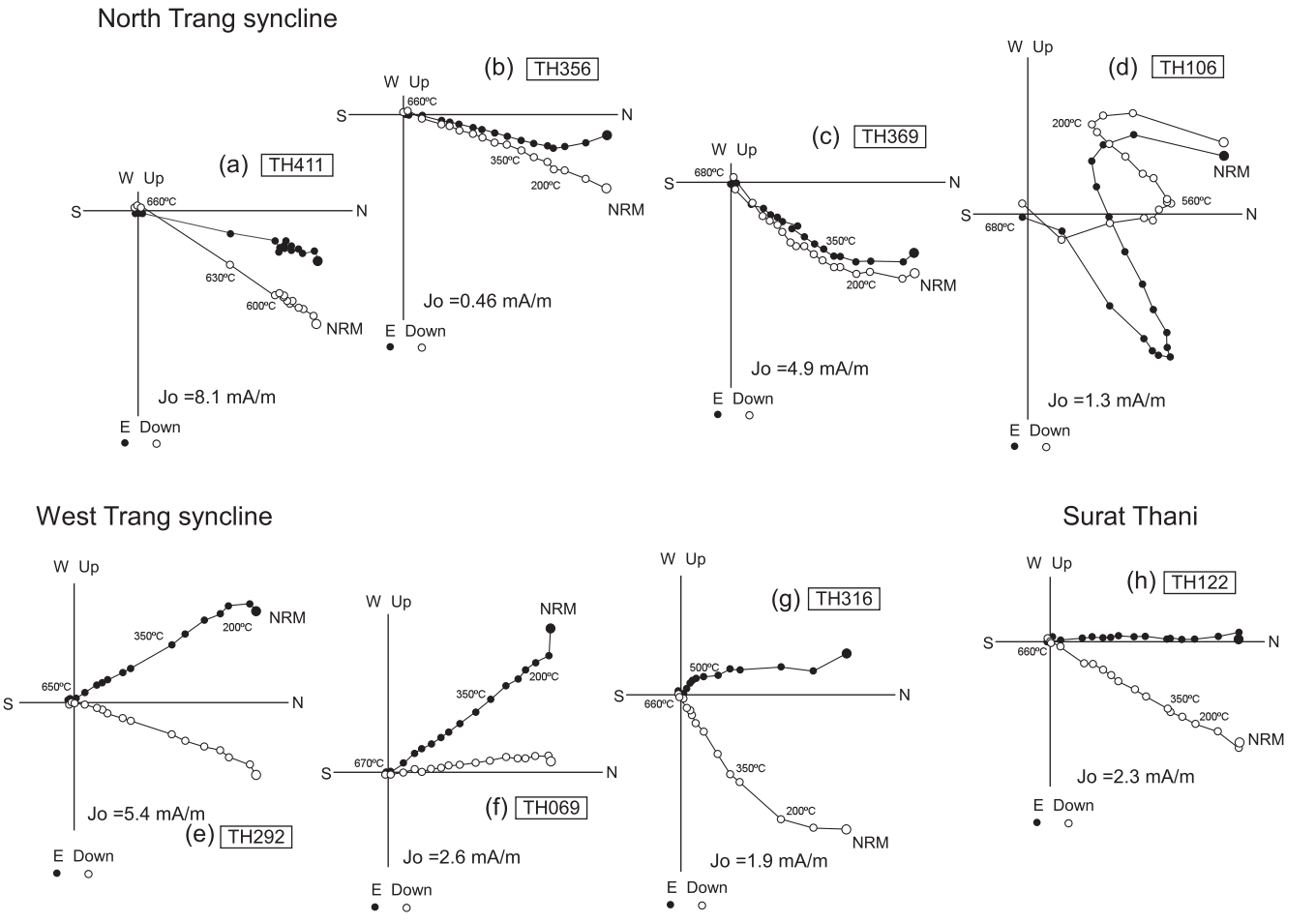


Fig. 4. Vector end-point diagrams of the representative specimens in geographic coordinates from North Trang Syncline (a, TH41; b, TH35; c, TH36; d, TH10), the West Trang Syncline (e, TH29; f, TH06; g, TH31) and the Surat Thani locality (h, TH12) after thermal demagnetization. Solid (open) symbols are projections onto horizontal (vertical) plane.

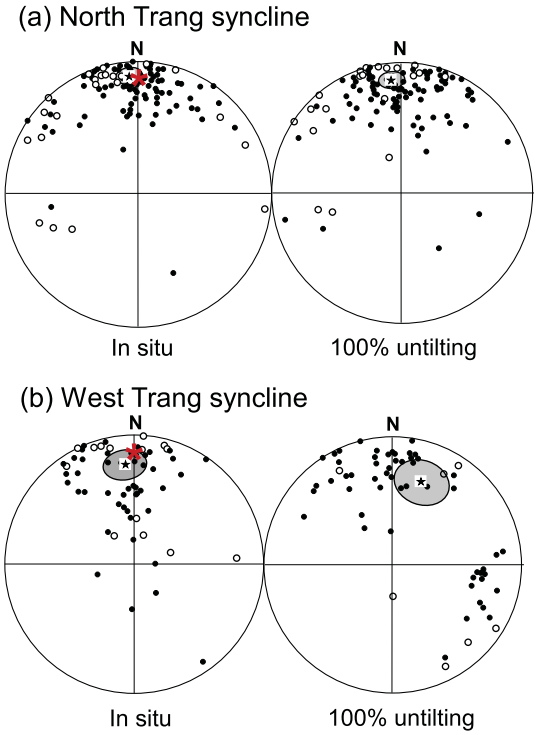


Fig. 5. Equal-area projections of the site mean directions (circles) for low-temperature component before and after tilt correction for (a) the North Trang Syncline and (b) the West Trang Syncline. Mean direction for each locality is shown by star with 95% circle of confidence (shaded). The asterisk represents the present axial geomagnetic dipole field. Solid and open symbols correspond to projection on lower and upper hemisphere, respectively.

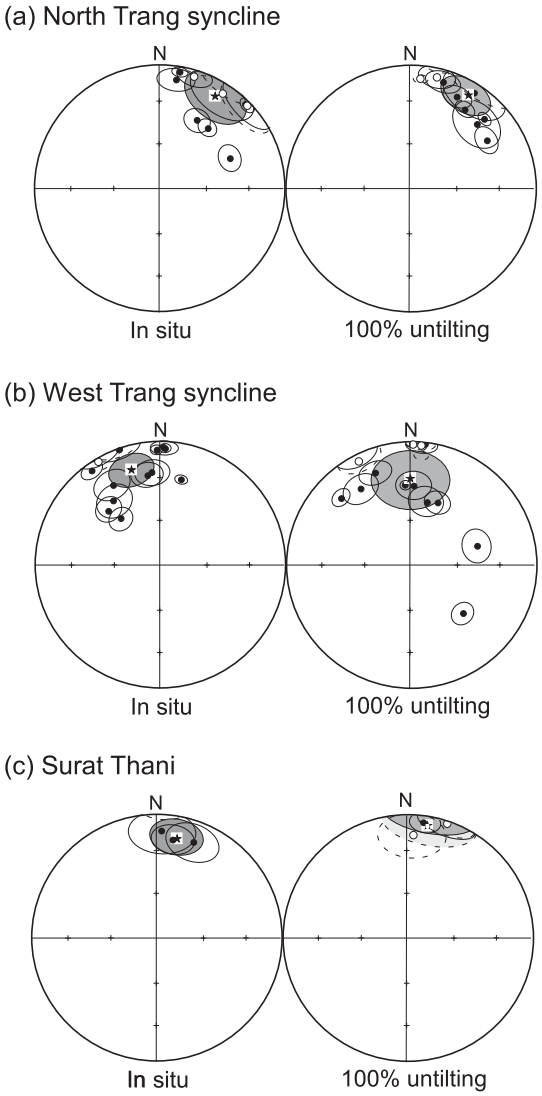


Fig. 6. Equal-area projections of the site mean directions (circles) for high-temperature component with 95% confidence limit before and after tilt correction from (a) North Trang Syncline, (b) West Trang Syncline and (c) Surat Thani locality. Mean direction for each locality is shown by star with 95% circle of confidence (shaded). Solid and open symbols correspond to projection on lower and upper hemisphere, respectively.

in geographic coordinates, and $D = 282.9°$, $I = -29.2°$, $\alpha_{95} = 24.6°$ in stratigraphic coordinates). Fold tests (Enkin, 2003; McFadden, 1990) applied to these four sites appeared inconclusive. Negative inclination values from these directions suggest that their respective magnetizations were probably acquired during a reversed polarity period. This component of magnetization can be taken into consideration for any future paleomagnetic endure in the study area because of its enigmatic direction.

5.2.2. The West Trang Syncline

Characteristic directions calculated from single-component magnetization of this locality appear parallel to in situ directions of the high-temperature component isolated from two-component magnetization of the Lam Thap and Klon Min formations. Therefore, mean direction is calculated on the basis of data from 13 sites (Table 1 and Fig. 6b). A precision parameter decreases from 11.6 to 4.2 upon tilt correction, and the DC-tilt test (Enkin, 2003) gives negative result. The McFadden's (1990) Fold test also proved negative at 95% confidence level, where a calculated value (ξ_1) of 0.1777 is obtained in the geographic coordinates and 5.382 in the stratigraphic coordinates, while a critical value (ξc) of 4.200 is estimated at 95% confidence level. These parameters thus indicate that paleomagnetic direction in the Middle–Late Jurassic to Late Cretaceous rocks of the West Trang Syncline is post-folding in origin. This is supported by directions in stratigraphic coordinates of the above 13 data set (Fig. 6b) which are dependent of the bedding attitude as observed in those of low-temperature component (Fig. 5b). It is, therefore, concluded that mean direction in geographic coordinates from all 13 sites ($D = 342.8°$, $I = 22.3°$, $k = 11.6$, $\alpha_{95} = 12.7°$) is considered as a characteristic direction for this area.

According to geological observations, synclinal structures in the Trang Group were formed prior to deposition of the Tertiary rocks (Kanjanapayont et al., 2002), so the acquisition of post-folding magnetization observed in the West Trang Syncline can be placed between the Late Cretaceous and Tertiary.

The unblocking and Curie temperatures observed in the rocks of West Trang Syncline are relatively lower than those of the North Trang Syncline. Based on these observations, it seems a reasonable assumption that carrier of magnetization in the West Trang Syncline is secondary hematite of CRM origin, which brought complete overprinting onto primary magnetization in the studied rocks (Dekkers and Linssen, 1989; Deng et al., 2007).

5.2.3. The Surat Thani area

As shown in Fig. 6c, mean direction obtained from the samples of Surat Thani area (including both the single and two-component magnetizations) show northerly declination and shallow inclination ($D = 11.1°$, $I = 19.0°$, $k = 78.4$, $\alpha_{95} = 14.0°$ in geographic coordinates, and $D = 10.6°$, $I = -4.9°$, $k = 33.9$, $\alpha_{95} = 21.5°$ in stratigraphic

Table 2
Paleozoic to Cenozoic paleomagnetic results from Sundaland Block. Lat. and Long. are for latitude and longitude, respectively. Age: P, Permian, Tr, Triassic, J, Jurassic, J1, Lower Jurassic, J2, Middle Jurassic, J3, Upper Jurassic; K, Cretaceous, K1, Lower Cretaceous; K2, Upper Cretaceous, T, Tertiary, Tpal, Paleogene, E, Eocene, M, Miocene. N, number of sites used for paleomagnetic calculations. Dec. and Inc. are declination and inclination, respectively. α_{95} and A_{95} are radii of cone of 95% confidence about the mean direction and the VGP, respectively. Diagramatic presentation of the VGPs is shown in Fig. 7.

Locality			Age	*N*(*n*)	Tilt correction	Fold test	Observed direction			VGP			References
Name	Lat. (°N)	Long. (°E)					Dec.(°)	Inc.(°)	α_{95} (°)	Lat.(°N)	Long.(°E)	A_{95} (°)	
Sundaland Block													
Peninsular Thailand													
Thung-Yai Red beds	8.38	99.4	J3–K2	1(23)	Corrected		32.6	18.8	3.9				Bhongsuwan and Ponathong (2002)
North Trang Syncline	8.1	99.4	J2–J3	9	Corrected	Positive	90.91	12.2	13.9	59.1	190.9	10.1	This study
(Intermediate-temperature component	8.1	99.4		4	In situ		96.6	18.0	25.8				This study
Krabi basin	8.0	99.0	20–40 Ma	71(*n*)	Magnetostratigraphy		12.3	15.5	4.7				Benammi et al. (2001)
West Trang Syncline	7.5	99.4	Post-K	13	In situ		342.8	22.3	12.7	72.6	24.3	9.8	This study
Peninsular Malaysia													
Segamat (basalt flow)	2.5	102.8	K2-Tpal	4	In situ		306.0	37.3	9.5	34.4	36.5	8.6	Richter et al. (1999)
Kuantan (diolite dike)	2.8–5.1	103.2–103.5	K2-T(?)	34	In situ		332.8	36.6	5.4	58.8	47.4	4.8	Richter et al. (1999)
Singapore (diolite dike)	1.4	104.0	K(?)	(15)	In situ		340.7	18.5	5.4	69.2	37.5	5.8	Richter et al. (1999)
Tembeling (red beds) (woTK-7)	3.5	101.8–103.0	J3-K1	13	In situ		325.5	28.1	8.8	54.3	325.52	8.18	Richter et al. (1999)
Main range Granite	3.0–6.4	99.8–102.4	Tru	7	In situ		42.5	50.0	12.4				Richter et al. (1999)
Permian silicic igneous rocks	1.4–1.7	104.2	P	4	In situ		26–66	−5∼39					Richter et al. (1999)
Borneo													
Telupid	5.7	117.1	K	6(48)	Corrected		313.3	−9.1	27.4				Schmidtke et al. (1990)
Bau Limestone Formation (Bau)	1.4	110.2	J–K	5(36)	In situ		301.6	11.5	14.1				Schmidtke et al. (1990)
Bau Limestone Formation (Penrissen)	1.4	110.2	J3-K1	2(17)	Corrected	Positive	272.0	−8.6	5.3				Schmidtke et al. (1990)
Bau-Lund Rd. (Padawan F.)	1.2–1.5	109.9–110.3	J3-K2	2(15)	Corrected		267.9	−16.0	7.1				Schmidtke et al. (1990)
Silantek Formation	1.1	110.9–111.5	E-M(?)	3(22)	In situ		318.7	24.3	13.4				Schmidtke et al. (1990)
Kedadan Formation	1.1	110.4	J3	3(21)	Corrected		311.1	−2.1	22.7				Schmidtke et al. (1990)
Kalimantan Tenguwe	0.7	110.1	J	11(40)	Corrected		267.1	2.6	6.9				Fuller et al. (1999)
W. Kalimantan-Schwaner Mtns.	1.5S	110.0	K	37(37)	In situ		311.7	0.3	7.8				Haile et al. (1977)
W. Kalimantan-Schwaner Mtns.	1.2–1.4S	110.0	K	1(5)	In situ		280.1	−5.6	24.5				Haile et al. (1977)
Batulicin	3.4S	115.9	K	1(5)	In situ		321.0	−13.0	14.4				Fuller et al. (1999)
East Sulawesi													
Biru	5.0S	120.2	Tpal-Lower M	9	In situ		315.2	1.3	19.6				Sasajima et al. (1980)
Dera Valley	4.75S	119.7	J3-K1	2(8)	Corrected		325	−5	7.2				Haile (1978)

coordinates). Since, the sampled sites are located over a monoclinal structure; a second definition (ξ_2) of the McFadden's method (1990) is applied, which gave a negative fold test at 95% confidence level. As a result of this test, a calculated value (ξ_2) of 1.2550 is obtained in the geographic coordinates and 2.502 in stratigraphic coordinates, while a critical value (ξc) of 2.076 is calculated at 95% confidence level. These parameters thus suggest that remanent magnetization in the rocks of Surat Thani area is post-folding in origin. Because of its similarity with axial geomagnetic dipole field ($D = 0°$, $I = 18.5°$) of the study area, an age comparable with Brunhes chron is presumed for the acquisition of this magnetization.

6. Discussion

6.1. Declination anomalies in the Trang area

The most striking aspect of this paleomagnetic study from Trang area is the declination anomalies in its western and northern parts, i.e., a northeasterly declination ($D = 31.1°$) from the Middle to Late Jurassic of North Trang locality and northwesterly declination ($D = 342.8°$) of the Late Cretaceous to Tertiary age from West Trang locality (Fig. 6). A change in declination from easterly to westerly position cannot be explained by the Eurasian APWP, because no CCW deflection is detected when expected values of declination for Trang area is calculated for Middle–Late Jurassic and Tertiary with respect to APWP of the SCB (Besse and Courtillot, 1991; Enkin et al., 1992; Yokoyama et al., 2001). This change in declination between different parts of the study area can rather be attributed to tectonic rotation in the Trang area.

Comparison between the observed and expected declinations (with respect to APWP of the SCB) suggest that the North Trang locality was subjected to CW tectonic rotation after the Middle to Late Jurassic, while the West Trang locality experienced CCW tectonic rotation after the Late Cretaceous. Geological observations from the study area indicate no large structural (faulted) boundary between the West and North Trang localities (DMR, 1999). The opposite sense tectonic rotations in the Tang area can thus be treated in time rather than in space, where the post-Middle to Late Jurassic CW rotation was followed by the Late Cretaceous to Tertiary CCW rotation.

6.2. CCW rotation of the Trang area and the southern Sundaland Block

With the help of this and previous paleomagnetic studies, timing and magnitude of CCW tectonic rotation in the Trang area is evaluated. A cessation in CCW rotation is inferred from northerly paleomagnetic direction ($D = 12.3°$, $I = 15.5°$, $k = 14$, $\alpha_{95} = 4.7°$ after tilt correction) reported from the Tertiary sedimentary rocks of the Krabi Basin (Benammi et al., 2001). Paleontological investigations from the area (Chaimanee et al., 1997; Chaodumrong and Chaimanee, 2002) proposes late Eocene-middle Miocene age for the continental deposition in the Krabi Basin, while an age of 31–34 Ma has been assigned to it by magnetostratigraphic correlation (Benammi et al., 2001). The CCW rotation thus took place between Late Cretaceous-Tertiary and Late Eocene-middle Miocene. Compared with 40 Ma pole position expected from APWP of the SCB (Enkin et al., 1992), magnitude of $24.5° \pm 11.5°$ is obtained as a minimum estimate of CCW rotation for Trang area between the Late Cretaceous and Middle Miocene, but without any latitudinal displacement ($-2.7° \pm 8.4°$).

This CCW rotation can fairly be regarded as northward extension of the same sense rotation experienced by the Peninsular Malaysia. As listed in Table 2, a range of CCW deflected declinations ($D = 306.0–332.8°$) have been reported from Late Jurassic to Early Tertiary rocks distributed between 1.4 and 5.2°N latitude in the Malaysian territory (Richter et al., 1999) (Table 2). The Jurassic to Cretaceous red beds of the Tembeling Formation, which are comparable to red beds of the Trang Group (Huchison, 1989), also reveals CCW declination ($D = 325.5°$) of Post-folding origin. Post-Jurassic Paleomagnetic poles from Trang area and the Malaysian Peninsula are shown in Fig. 7. Paleomagnetic pole from Trang area (72.6°N, 24.3°E) falls neighboring to poles from the Peninsular Malaysia (between 34.4°N, 36.5°E and 69.2°N, 37.5°E), suggesting an occurrence of CCW tectonic rotation in both these areas coherently.

This CCW tectonic rotation of the Thai–Malay Peninsula is a part of Cenozoic CCW rotation occurred in the South Sundaland Block (including Borneo, southern Sulawesi and Celebes sea) (Table 2). Similar trend of declination ($D = 270–320°$) has been observed in the Jurassic to lower Miocene rocks of southern Sundaland Block, for which the middle Miocene has been assigned as an acquisition age (Haile, 1978; Sasajima et al., 1980; Schmidtke et al., 1990; Shibuya et al., 1991; Richter et al., 1999; Fuller et al., 1999).

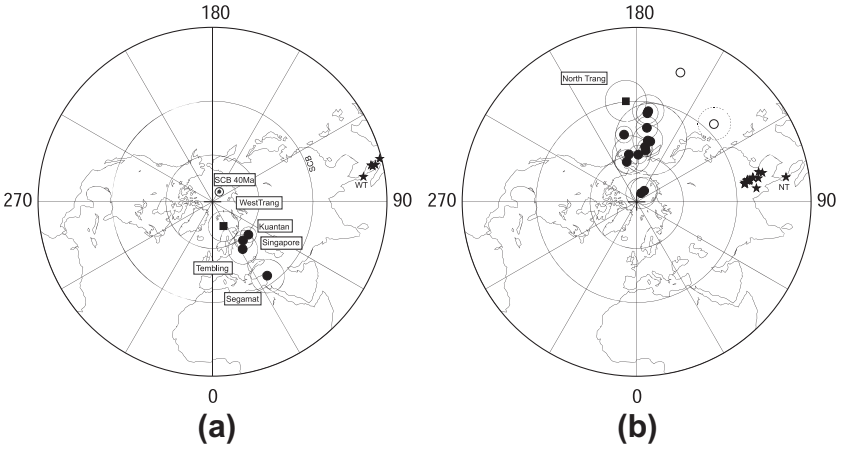


Fig. 7. Stereographic projection of the Jurassic to Paleogene paleomagnetic poles from Thai–Malay Peninsula together with Shan-Thai and Indochina blocks and their 95% confidence circles. Poles obtained from Middle–Late Jurassic to Cretaceous rocks of Trang area are shown by squares, while those from other areas are indicated by circles. (a) A paleomagnetic pole from West Trang Syncline (72.6°N, 24.3°E) together with Jurassic to Paleogene poles from Peninsular Malaysia (Segamat, Kuantan, Singapore and Tembling areas from Richter et al., 1999). (b) A paleomagnetic pole from North Trang Syncline together with Jurassic–Cretaceous poles from Shan-Thai and Indochina blocks (Yang and Besse, 1993; Takemoto et al., 2009). A revised pole for North Trang Syncline (34.6°N, 187.1°E) is calculated on the basis of revised paleomagnetic direction of $D = 55.6°$, $I = 12.2°$, assuming that this syncline was subjected to CCW tectonic rotation by 24.5° .

This widely observed CCW rotations from southern Sundaland Block are, however, in contrasts to CW rotation reported from the Indochina and Shan-Thai blocks (Fig. 1). Based on the present study, a boundary between the CCW and CW rotated terranes within the Sundaland Block is placed between the Trang area and the Khorat Basin, which can clearly be recognized by geomorphologic bending in the Malaysian Peninsula. Previously the Songkhla-Penang Fault in the Malaysian Peninsula has been proposed as a transition zone between differential rotational zones (Bunopas, 1982; Richter et al., 1999), however, the present study suggest a shift in this boundary up to 300 km towards north into Peninsular Thailand. A transition zone between these oppositely rotated blocks is nested by four different fault systems (including the Mae Ping, the Pagodas, the Panong and the Klong Marui faults), which run almost parallel to this zone (Fig. 1). As suggested by Hall (1996) and Morley (2002), these faults have played an important role in accommodating opposite sense rotational motion between the CCW rotated southern Sundaland Block and a CW rotated Indochina Block.

A characteristic tectonic feature of the CCW rotation in the southern Sundaland Block has probably been induced by a collision of Australian Plate with SE Asia. According to geological observations from the area (Lee and Lawver, 1995; Hall, 2002; Pubellier et al., 2003; Smyth et al., 2007), northern tip of the Australian Continent collided with Sundaland Block between 30 and 20 Ma. Timing of CCW rotation in the Sundaland Block is contemporaneous to this collisional event (Fuller et al., 1999). This is in analogy to CW rotation of the Shan-Thai and Indochina terranes (parts of the northern Sundaland Block), which has been attributed to extrusion tectonics in East Asia as a result of Indian Plate indentation (Tapponnier et al., 1982; Replumaz and Tapponnier, 2003). Tectonics in the southern part of Sundaland Block as a result of its collision with Australian continental margin can possibly be considered as a mirror image of what happened around eastern Himalayan syntaxis after India–Asia collision (Fig. 8).

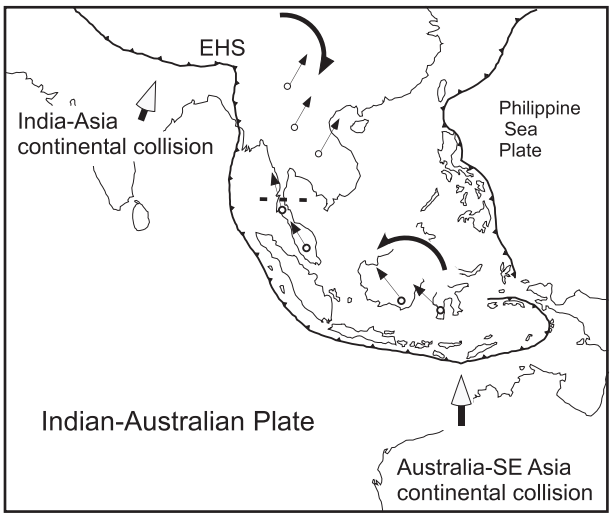


Fig. 8. Tectonic model of differential tectonic rotation in the Sundaland Block. Whilst, CW rotation has been induced around eastern Himalayan syntaxis (EHS) during the collision of India with Asia, southern part of the Sundaland Block experienced CCW rotation as a result of Australian Plate collision with SE Asia. These rotational events are the mirror images of each other, and a boundary separating them is located between the Khorat Basin and the Trang area. Small arrows in the model represent observed declinations of the Cretaceous to Jurassic age from Shan-Thai, Indochina, West Trang Syncline, Peninsular Malaysia, Kalimantan and Sulawesi of the Sundaland Block.

6.3. CW deflected declination in the Trang area

CW tectonic rotation occurred both in the Peninsular Malaysia as well as in the Trang area. As reported by Richter et al. (1999), the Permian to Triassic rocks in the Malaysian Peninsula preserved the CW deflected declinations, i.e., $D = 42.5°$, $I = 50.0$, $\alpha_{95} = 12.4°$ from Late Triassic granites; $D = 26–66°$ from Permian to Triassic silicic igneous rocks; and $D = 26–36°$ from remagnetized Paleozoic carbonates. Since, the CW paleomagnetic declination obtained from Middle–Upper Jurassic rocks of the Trang area (this study) is recognized as pre-folding in origin, we consider it as a first reliable discovery in the Mesozoic from Thai–Malay Peninsula.

Ferrari et al. (2008) have reported that the process of amalgamation between the Sibumasu and Indochina blocks accomplished during the Middle Triassic to Earliest Jurassic along the Bentong–Raub fracture zone (a part of Peninsular Malaysia). Since, CW rotation in the Thai–Malay Peninsula has been recognized to have occurred after the amalgamation, post-Middle to Late Jurassic time can thus be assigned to this event. This implication is supported by the fact that CW paleomagnetic declinations have been observed on both sides of the Bentong–Raub fracture zone in the Malaysian Peninsula (Richter et al., 1999).

Based on these evidences, it can be suggested that the Thai–Malay Peninsula have experienced a coherent rotational motion together with Shan-Thai and Indochina blocks before it received CCW rotation in the Cenozoic. Taking into consideration this Cenozoic CCW rotation of 24.5° in the southern part of Sundaland Block, paleomagnetic declination of the Middle to Late Jurassic rocks from North Trang locality is re-adjusted to $D = 55.6°$. Based on this new estimate, a revised paleomagnetic pole of 34.6°N, 187.1°E (Fig. 7) for Trang area falls close to Jurassic–Cretaceous poles from Shan-Thai and Indochina blocks (Yang and Besse, 1993; Takemoto et al., 2009). This new estimation further suggests that the Thai–Malay Peninsula and the Shan-Thai–Indochina blocks have probably experienced same magnitude of CW rotation after the Late Jurassic. Compared with paleomagnetic direction expected from the Jurassic pole of the SCB (Yokoyama et al., 2001), the Thai–Malay Peninsula is considered to have experienced southward displacement of 15.4° ± 9.7° after the Jurassic, which is concordant with post-Cretaceous southward displacement (7.1° ± 2.2°) of the Shan-Thai and Indochina blocks (Takemoto et al., 2009). These facts thus suggest similar behavior for Thai–Malay Peninsula and Shan-Thai–Indochina blocks after the Jurassic–Cretaceous. Since, the age of this CW rotation in the Thai–Malay Peninsula is unclear; it cannot be ascribed to Cenozoic collision of India with Asia.

7. Conclusions

The present study focuses on paleomagnetic investigations conducted on the Middle–Late Jurassic to Cretaceous red sandstones (the Khlong Min and Lam Thap formations) from around the Trang city (7.6°N, 99.6°E), southern part of the Thailand.

(1) A post-folding remanent magnetization obtained from West Trang Syncline shows westerly deflected declination ($D = 342.8°$, $I = 22.3°$, $\alpha_{95} = 12.7°$), indicating CCW tectonic rotation of about 28° with respective to the SCB between the Late Cretaceous and Middle Miocene. This post-Cretaceous rotational motion in the Trang area is a part of large scale CCW rotation experienced by southern Sundaland Block as a result of Australian Plate collision with SE Asia.
(2) In contrast to West Trang Syncline, a pre-folding characteristic magnetization ($D = 31.1°$, $I = 12.2°$, $\alpha_{95} = 13.9°$) is obtained from Middle to Late Jurassic Khlong Min Formation of the North Trang Syncline, which shows CW tectonic rotation of the studied syncline.

(3) A sequence of two opposite trend tectonic rotations explains the paleomagnetic characteristic of Trang area together with Peninsular Malaysia, where the Cenozoic CCW rotation was preceded by post-Jurassic CW rotation.
(4) Comparison with previously reported paleomagnetic data from adjacent areas suggests that the Thai–Malay Peninsula (including the Trang area) underwent a CW tectonic rotation together with Shan-Thai and Indochina blocks after the Late Jurassic.

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